

1. Explain what is data science? How can we discover knowledge from the data?

→ Data science is an interdisciplinary field that uses scientific methods, statistical techniques, machine learning algorithms, and computational tools to collect, process, analyze and interpret data in order to extract meaningful insights and support decision making.

→ The primary goal of data science is to transform raw data into useful information and actionable knowledge.

⇒ Knowledge Discovery in Database (KDD) is the process of extracting useful, understandable, and previously unknown patterns from large datasets. It helps organizations convert raw data into valuable knowledge.

→ The KDD process consists of the following stages:-

1> Data Selection:-

- Selection is the processing of identifying and retrieving relevant data from various data sources such as DB, data warehouse, & data streams.

2> Data preprocessing:-

- preprocessing involves cleaning and preparing the selected data for analysis.
- Real-world data is often incomplete, inconsistent,

noisy or duplicated. Such issues can negatively affect the quality of the result- so they must be corrected before further analysis.

3. Data Transformation:-

- Transformation converts the cleaned data into a suitable format for data mining. Raw data may not always be appropriate for analysis so various transformation techniques are applied to improve its usefulness.

- Common techniques.

- Normalization, Standardization, Discretization, Aggregation, Feature construction.

4. Data mining:-

- Data mining is the core step of the KDD process.

- In this stage, algorithms and statistical techniques are used to discover hidden patterns, relationships trends, or knowledge from the transformed data.

- Common data mining tasks:-

- Classification, Regression, clustering, Association Rule Mining, Anomaly Detection.

5. Evaluation:-

- Evaluation is the process of assessing the discovered patterns or models to determine whether they are useful, accurate and meaningful.

- Not all patterns found during data mining are valuable, therefore, evaluation helps identify the most relevant know

2. Differentiate between structured and unstructured data with eg:-

→ feature	structured Data	unstructured Data.
• Definition:-	- Data that follows a fixed schema and organized format	- Data that has no predefined format or structure.
• Organization:-	- Highly organized in rows and columns	- No specific organization or predefined model
• Storage :-	- stored in relational DB (RDBMS)	- stored in data lakes, file systems or NoSQL database.
• Data Model:-	- Requires predefined schema before storage	- No fixed schema required (schema-on-read).
• Processing:-	- Easy to process using SQL and standard tools	- Requires advanced techniques such as AI, ML, NLP or CV.
• Querying:-	- Fast, efficient and straightforward querying	- complex and often requires specialized tools for analysis.
• Flexibility:-	- less flexible due to rigid structure	- Highly flexible & adaptable to different formats.
• E.g:-	- Databases, spread-sheets, financial records	- Text, images, videos, emails, social media content.

3. Explain the EDA process in Data Science.

- Exploratory Data Analysis (EDA) is the process of examining, summarizing and visualizing data to understand its main characteristics before applying statistical models or machine learning algorithms. It helps data scientists discover patterns, trends, relationships and anomalies within a dataset.
- EDA is an important step in the data analysis pipeline because it provides insights into the structure and quality of data, helping analysts make informed decisions about data preprocessing and modeling.

⇒ Steps in the EDA process.

1. → Data understanding:-

- Examine the dataset, variables, data types, and overall structure
- Identify numerical and categorical attributes.

2. → Data cleaning:-

- Handle missing values, duplicates, inconsistencies, and outliers.
- Improve the quality of data for analysis.

3. → Distribution Analysis.

- Study how data values are distributed using histogram and frequency distributions.
- Helps identify patterns, skewness & spread of data.

4> Relational Analysis-

- Analyze relationships between variables using correlation and cross-tabulation.
- Helps understand dependencies and associations among variables.

5> Data visualization.

- Use charts and graphs such as histograms, bar charts, and scatter plots.
- Make patterns and trends easier to understand.

6> Insight Generation:-

- Draw meaningful conclusions from the analyzed data.
- Helps in decision-making and model building.

4. What is data cleaning? Explain different ways of handling missing data with e.g.

→ Data cleaning is the process of detecting, correcting, or removing incomplete, incorrect, duplicate or inconsistent data from a datasets.

→ It is important step in data preprocessing because real-world data is often noisy and incomplete.

→ The main purpose of data cleaning is to improve the quality and accuracy of data before analysis.

⇒ Handling missing Data.

→ Missing data occurs when some values are not available in the dataset.

↳ Different methods are used in handling missing data.

1. Ignore or remove the missing Record.

- Row containing missing values are deleted

e.g.

Name	Marks		Name	Marks
A	80	⇒	A	80
B	-		C	75
C	75			

- Advantage:- Simple & easy method.

- disadvantage:- loss of useful information if many records are removed.

2. Fill missing values manually.

- The missing values are entered manually using domain knowledge.

eg If a teacher knows that a student scored 70 marks, the missing value can be filled manually.

- Advantage:- More accurate if expert knowledge is available.
- disadvantage:- Time consuming for large datasets.

3. Replace with Global Constant.

- Missing values are replaced with a fixed value such as:

- unknown, 0, N/A

eg

Name	City		Name	City
Ram	Kathmandu	⇒	Ram	Kathmandu
Hari	—		Hari	Unknown.

- Advantage:- Easy to implement.
- disadvantage:- may reduce data quality.

4. Replace with Mean, Median or Mode

(a) Mean Method.

- used for numerical data.

eg:-

Marks:- 60, 70, —, 80

$$\text{Mean} = (60 + 70 + 80) / 3 = 70.$$

∴ Missing value is replaced with 70.

(b) Median Method

- used when data contains outliers.

eg:-

10, 12, —, 14, 100

Median = 13,

∴ Missing value is replaced with 13

(c) Mode Method

- used for categorical data.

eg! Red, Blue, Red, — Green.

Mode = Red,

∴ Missing value is Replaced with Red.

- Advantage:- Maintains data set size.

- disadvantage:- can slightly reduce accuracy.

5) Replace using prediction methods.

- Advanced techniques like

- Regression

- Decision tree

- Machine learning algorithms.

are used to predict missing values.

e.g:-

If salary depends on experience and age, the missing salary can be predicted using those attributes.

- Advantage:- More accurate.

- Disadvantage:- Complex and computationally expensive.

5. > Why is Data Discretization Necessary in Data Science?
Difference b/w Standardization & Normalization.

→ Data Discretization is the process of converting continuous numerical data into a finite number of intervals or categories (bins).

→ It simplifies data by reducing the no of distinct value and makes analysis easier.

⇒ Data Discretization is Necessary because:-

a) Reduces Data Complexity.

- large ranges of numerical values are converted into meaningful categories.

b) Improves Data Understanding

- Categorized data is easier to interpret & analyze.

c) Reduces Noise

- Small fluctuations in data are minimized.

d) Improves performance of some Algorithms.

- certain machine learning algo works better with 'categorical' data.

e) Facilitates pattern discovery.

- makes hidden trends and relationship easier to identify.

eg

Age data:- 18, 22, 25, 30, 35, 40, 45.
After discretization.

Age	category
18-25	Young
26-40	Adult
41-60	Middle-aged.

⇒ Difference b/w Standardization and Normalization:-

↳ Both techniques are used to scale numerical data but they use different approaches.

Standardization

- Transforms data so that it has mean = 0 and standard deviation = 1

- values are centered around zero
- Useful when data follows a normal distribution
- Can produce negative values
- less affected by outliers than normalization

$$Z = \frac{X - \mu}{\sigma}$$

original value
mean.
standard deviation.

Normalization

- scales data to a fixed range, usually 0 to 1
- values lie within a specified range.
- Useful when features have different scales.
- Usually produces values between 0 & 1
- More sensitive to outliers.

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

6. Explain what is regression? Determine the missing value using linear regression.

→ Regression is a statistical and machine learning technique used to study the relationship between a dependent variable and one or more independent variable.

⇒ It helps predict the value of a continuous variable based on the values of other variables.

⇒ Regression finds a mathematical relationship b/w variable and uses the relationship to make predictions.

Here,

x_1	60	62	67	70
x_2	22	25	24	26
y	140	155	159	?

→ Using Multiple Linear Regression, assume the model,

$$Y = a + b_1 x_1 + b_2 x_2. \quad \text{--- (1)}$$

now,

For value, 60, 22, 140.

$$140 = a + b_1(60) + b_2(22). \quad \text{--- (2)}$$

For value, 62, 25, 155.

$$155 = a + b_1(62) + b_2(25). \quad \text{--- (3)}$$

for

for value, 67, 24, 159,

$$159 = a + b_1(67) + b_2(24) \quad \text{--- (4)}$$

now,

~~eqs~~ subtracting
solving eq (2), (3) & (4)
we get,

$$a = -42$$

$$b_1 = 1.58$$

$$b_2 = 3.94$$

then eq (1) becomes,

$$Y = -42 + (1.58) \cdot x_1 + (3.94) \cdot x_2$$

now, For $x_1 = 70$, $x_2 = 26$.

$$Y = -42 + 1.58 \times 70 + 3.94 \times 26$$

$$= 171.04$$

$\therefore$ The missing value of Y using linear regression is.

$$Y = 171.04 \approx 171$$

7. Explain How you Determine the Best Fit line.

→ A best fit line, also known as a regression line, is a straight line that best represents the relationship between an independent variable (x) and a dependent variable (y).

→ It is used in linear regression to predict values & analyze trends in data.

→ The eqⁿ of best fit line is.

$$y = a + bx.$$

Where,

y = predicted value (dependent variable).

x = independent variable.

a = Intercept.

b = slope of the line.

⇒ Determining the Best Fit line

→ The best fit line is determined using the least square method.

⇒ principle of least square method.

↳ The regression line is chosen so that the sum of the squared difference b/w the actual & predicted values is minimum.

$$\text{i.e. Error} = \sum (y_A - \hat{y}_p)^2$$

$y_A \rightarrow$ Actual value

$\hat{y}_p \rightarrow$ predicted value.

$\Rightarrow$ Steps to Determine the Best Fit Line.

1. $\rightarrow$ plot the Data

- Represent the observation on a scatter plot.

2. $\rightarrow$ calculate the slope (b).

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$

3. $\rightarrow$ Calculate the intercept.

$$a = \bar{y} - b\bar{x}$$

4. $\rightarrow$ Form the Regression Eqⁿ.

$$y = a + bx$$

5. $\rightarrow$ Evaluate the Fit.

$\rightarrow$ The line with the smallest sum of squared error is considered the best fit line.

8. Differentiate between categorical and Numerical Data with eg:-
Explain which types of Data visualization is suitable for Numerical Data and categorical Data.

Numerical Data:-

→ Numerical data consists of values that represent quantities and can be measured or counted.

→ Mathematical operations such as addition, subtraction, multiplication and averaging can be performed on numerical data.

→ eg:- Age of students, salary of employee, Height & weight, temperature, etc.

→ Types of Numerical Data:-

1. Discrete Data:- Countable value. (eg no of students in class)

2. Continuous Data:- Measurable value (eg height, weight, temperature).

→ Categorical Data:-

- Categorical data consists of labels or categories that describes the characteristics of an object.

- Mathematical operations cannot be meaningfully performed on categorical data.

→ eg:- Gender (Male, Female), Blood Group (A, B, AB, O), Marital status, product category, education level etc.

→ Types of categorical data:-

↳ Nominal Data:- categories without any order
(eg blood group)

2) Ordinal Data:- Categories with a meaningful order (eg low, medium, High).

⇒ Numerical Data Categorical Data.

Q) Represents quantities or measurements

b) Mathematical operations can be performed

e.) Values are numeric

d) Can be discrete or continuous

e.g. eg:- Age, salary, Height

a.) Represents labels or groups.

b) Mathematical operations cannot be performed.

c) values are categories or labels.

d) can be nominal or ordinal.

e.g. Gender, Blood Group, Department.

⇒ Suitable Visualization For Numerical Data.

→ Histogram

- shows the distribution of numerical values.

- Helps identify spread, skewness, & outliers.

e.g:- distribution of student's marks.

2.) Scatter plot.

- shows the relationship between two numerical variables.

eg:- Study hours vs. Mark.

3.) Box plot.

- Displays median, quartiles, & outliers.

eg:- salary distribution of employee.

4.) Line chart.

- Used to show trends over time.

eg:- Monthly sales revenue.

⇒ Suitable visualization for categorical data.

1.) Bar chart.

- Compares frequencies of different categories.

eg:- No of students in different department.

2.) Pie chart.

- Shows the proportion of each category.

eg:- Market share of different smartphone brands.

3.) Stacked or Grouped Bar chart.

- Compares multiple categorical groups.

eg:- product sales across different regions.

Q) What is power law? Differentiate between Normal & Binomial distribution.

→ power law is statistical relationship in which a small number of events occur very frequently while a large number of events occur very rarely. In other words, a few items have very large values whereas most items have very small values.

$$p(x) = C x^{-\alpha}$$

α = scaling exponent (typically $\alpha \geq 1$ for real world system).

C = normalization constant ensuring total probability sum to 1

As x increases, probability decreases polynomially rather than exponentially.

2) Characteristics of power law.

- Heavy-tailed distribution.
- few large events & ~~small~~ many small events.
- common in nature & social system.

Normal Distribution

- Continuous probability distribution
- Bell shaped & symmetric
- Defined by mean & standard deviation
- value can take any real no within a range
- used for measurements & natural phenomena
- eg:- Height, weight, IQ

Binomial Distribution

- Discrete probability distribution
- Based on success / failure outcome
- Defined by no of trials (n) & probability (p)
- values are whole no only
- used for counting successes in trials
- eg:- coin tosses, pass / fail, outcomes